

CATHERINE CHEN

she/her/hers ◇ Updated Oct 2026
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RESEARCH INTERESTS

interpretability/explainability, information retrieval, natural language processing, large language models, human-centered AI, domain-specific applications

EDUCATION

Brown University <i>Ph.D in Computer Science</i> <i>Sc.M in Computer Science</i> Committee: Carsten Eickhoff (Advisor), Ellie Pavlick, Avishek Anand (TU Delft)	Providence, RI, USA Sept 2021 - July 2026 May 2023
Wellesley College <i>B.A. in Computer Science and French</i>	Wellesley, MA, USA Aug 2015 - May 2019

ACADEMIC EXPERIENCE

Leiden University <i>Postdoctoral Researcher</i> Supervised by Suzan Verberne and Ilaria Tilidi (VU)	Leiden, The Netherlands Oct 2026 - Present
University of Amsterdam <i>Visiting Researcher at IR Lab</i> Hosted by Maarten de Rijke and Maria Heuss	Amsterdam, The Netherlands Feb - May 2025
University of Massachusetts, Amherst <i>Research Assistant</i> Supervised by David Jensen	Amherst, MA, USA June - Aug 2017

WORK EXPERIENCE

Centre for the Governance of AI <i>Summer Fellow</i> Designed a framework to operationalize explainability auditing, mapping regulatory transparency goals to granular explanation properties and existing XAI tools.	London, UK June - Aug 2025
Blue Cross and Blue Shield of Minnesota <i>Data Science PhD Intern, AI Models and Platform Team</i> Collaborated with healthcare professionals to develop an end-to-end pipeline leveraging LLMs using AWS services (Sagemaker, Bedrock, S3) for automated Behavioral Health patient case summarization.	Remote (from Providence, RI, USA) June - Sept 2024
FreeWheel, A Comcast Company <i>Software Engineer, Creative Team</i> - Implemented backend APIs in Golang with gRPC to ingest and manage advertising creatives stored in AWS S3/MySQL DB, supporting large-scale ad inventory workflows. - Constructed user interface components for a React-based web application within a service-oriented, containerized backend architecture using Docker and Kubernetes.	New York, NY, USA Aug 2019 - Aug 2021
PatientsLikeMe <i>Software Engineering Intern</i> Developed features for a large-scale web application using Ruby on Rails and React, supporting a personalized health platform for individuals with chronic conditions.	Cambridge, MA, USA June - Aug 2018

SELECTED PUBLICATIONS

**Denotes equal contribution*

- **Catherine Chen**, Maarten de Rijke, and Carsten Eickhoff. *From Interpretation to Intervention: Can Mechanistic Insights on Gender Sensitivity Inform Targeted Interventions?* In submission.
- Yumeng Wang, **Catherine Chen**, and Suzan Verberne. *RankSteer: Can Pointwise LLM Rankers Be Calibrated at the Representation Level?* To appear in EMNLP 2026.
- Yumeng Wang, Jirui Qi, **Catherine Chen**, Panagiotis Eustratiadis, and Suzan Verberne. *How Role-Play Shapes Relevance Judgment in Zero-Shot LLM Rankers*. ECIR 2026.
- Meng Lu*, **Catherine Chen***, and Carsten Eickhoff. *Pathway to Relevance: How Cross-Encoders Implement a Semantic Variant of BM25*. EMNLP 2025.
- Gregory Polyakov, **Catherine Chen**, and Carsten Eickhoff. *Towards Best Practices in Axiomatic Activation Patching in Information Retrieval*. SIGIR 2025.
- Andrew Parry, **Catherine Chen**, Carsten Eickhoff, and Sean MacAvaney. *MechIR: A Mechanistic Interpretability Framework for Information Retrieval*. ECIR 2025.
- **Catherine Chen**, Jack Merullo, and Carsten Eickhoff. *Axiomatic Causal Interventions for Reverse Engineering Relevance Computation in Neural Retrieval Models*. SIGIR 2024.
- **Catherine Chen** and Carsten Eickhoff. *Evaluating Search System Explainability with Psychometrics and Crowdsourcing*. SIGIR 2024.

MENTORSHIP

Master's Students

Meng Lu – Brown University → MIT PhD

July 2024 - Sept 2025

Kora Helm – University of Tübingen → University of Augsburg PhD

Sept 2023 - July 2024

Undergraduate Students

Pearl Zhang – Swarthmore College – with Prof. Jeff Huang

Jan - May 2023

Yuhan Wang – Smith College → Georgia Tech PhD – with Prof. Jeff Huang

Jan - May 2023

Lucas Weissman – Williams College – with Roma Patel

June - Aug 2022

Marwa Bouabid – Mount Holyoke – with Prof. James Tompkin

Jan - May 2022

TEACHING

Brown University

CSCI 1460: Computational Linguistics, *Graduate TA*

Sept - Dec 2024

Sheridan Center for Teaching and Learning, *Teaching Certificate I*

Sept - Dec 2023

NSF REU: AI for Computational Creativity, *Research Methods Study Group Leader*

June - Aug 2022, 2023

Wellesley College

CS 235: Theory of Computation, *Supplemental Instructor*

Sept - Dec 2018

INVITED TALKS

Peeking Under the Hood: Interpretability and Explainability in IR

European Summer School on Information Retrieval (ESSIR)

July 2025

Reverse Engineering Mechanisms of Relevance in Neural IR Models

University of Amsterdam

May 2025

Radboud University

May 2025

Leiden University

April 2025

Advancing Explainable Information Retrieval: Methods for Human-Centered Evaluation and Interpreting Neural IR Models

University of Glasgow

Sept 2024

Choosing Between Industry and Academia

Brown University, ExploreCSR: Diversity in Computer Systems

Apr 2023

Women in STEM Workshop: Pursuing a Career in CS

The Stemistry Project

Feb 2023

LEADERSHIP/SERVICE

Faculty-Grad Liason

Department of Computer Science

Brown University

Fall 2024 - June 2026

- Communicate with the department about graduate student affairs/concerns and coordinate student volunteer positions.

CS Department Representative

Graduate Student Council

Brown University

Sept 2022 - May 2024

- Attended monthly meetings to facilitate collective action on graduate student-related issues.

Graduate Student Orientation and Recruitment

Department of Computer Science

Brown University

Jan 2022 - Mar 2024

- Collaborated with department administrators to organize welcome events for admitted/incoming students.

Graduate Community Fellow

Graduate School

Brown University

Sept 2022 - May 2023

- Designed and implemented community-building initiatives for Women and Non-Binary Graduate Students to celebrate diversity, promote belonging and inclusion, and connect individuals across disciplines.

Co-Organizer

Workshop on Explainability in Information Retrieval (WExIR) @ SIGIR

2025, 2026

Tutorial on Mechanistic Interpretability @ ECIR

2026

New England NLP Meetup

2024

Conference Reviewer

Actionable Interpretability Workshop

2025

ACL Rolling Review (EMNLP '24 Outstanding Reviewer)

2024 - 2025

CHIIR

2026

ECIR

2024 - 2026

SIGIR

2024 - 2026

WSDM

2025

Panelist

Amplify: Students as Leaders: At Wellesley and Beyond

Oct 2024

Wellesley CS Club: Wellesley Careers in Tech Panel

Apr 2024

AlumConnect: Research at Wellesley

Feb 2024

AI/ML Panel at Hack @ Brown

Feb 2024

Moderator

Grad Student Discussion Panel for REU AI @ Brown

Aug 2023

Grad Student Discussion Panel for ExploreCSR AI @ Brown

Apr 2023

Women and Non-Binary Faculty Panel @ Brown

Mar 2023

Other Service

Max Planck Institute for Intelligent Systems PhD Application Reader

Nov 2024

Brown CS PhD Application Reader

Jan 2024

AWARDS

Research Mobility Fellowship, Brown University

Spring 2025

Graduate Community Fellow, Brown University

Fall 2022 - Spring 2023